PRACTICAL-1 : HADOOP INSTALLATIONS

Steps to Install Hadoop

• Install Java JDK 1.8

• Download Hadoop and extract and place under C drive

• Set Path in Environment Variables

• Config files under Hadoop directory

• Create folder datanode and namenode under data directory

• Edit HDFS and YARN files

• Set Java Home environment in Hadoop environment

• Setup Complete. Test by executing start-all.cmd

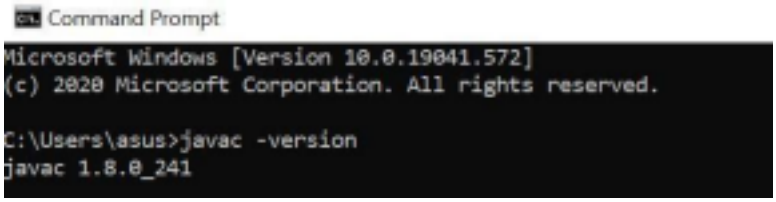
1. Install Java

https://www.oracle.com/java/technologies/javase-jdk8-downloads.htm

First check that jdk is 1.8 bcoz Hadoop support 1.8

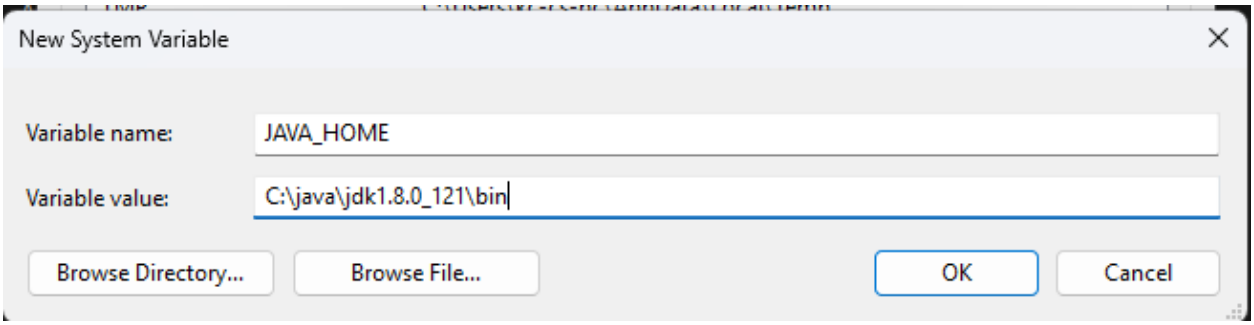
Downlord n install java 1.8

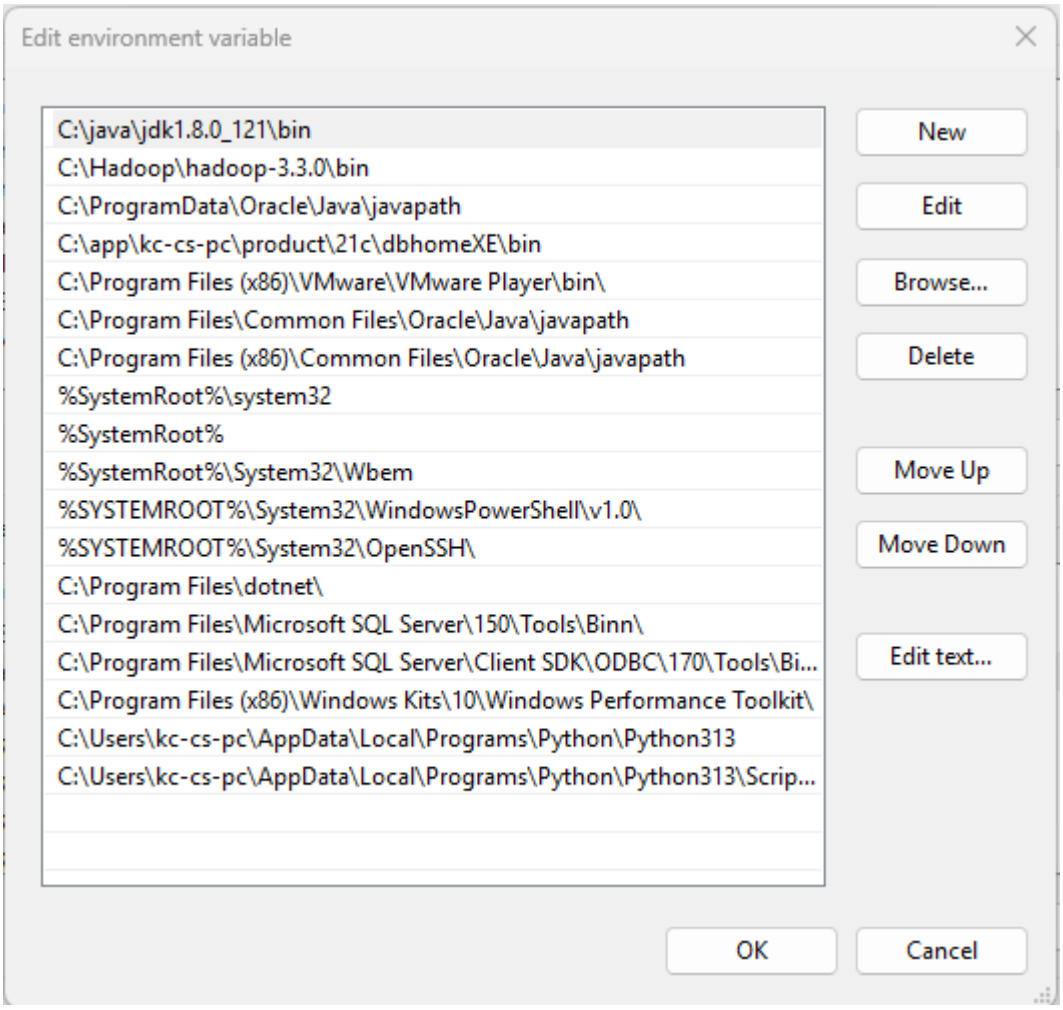
see version in cmd javac –version



create folder in C name java copy that jdk in that folder

Set the path JAVA_HOME Environment variable and set the path till bin





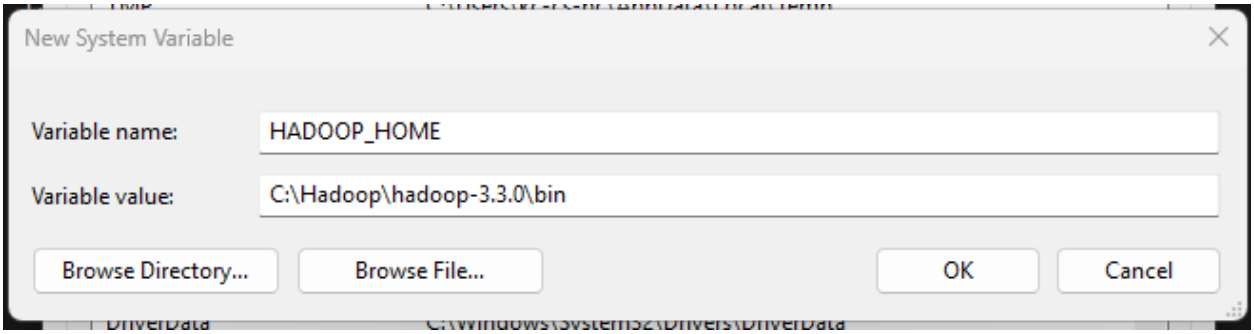
2. Download Hadoop

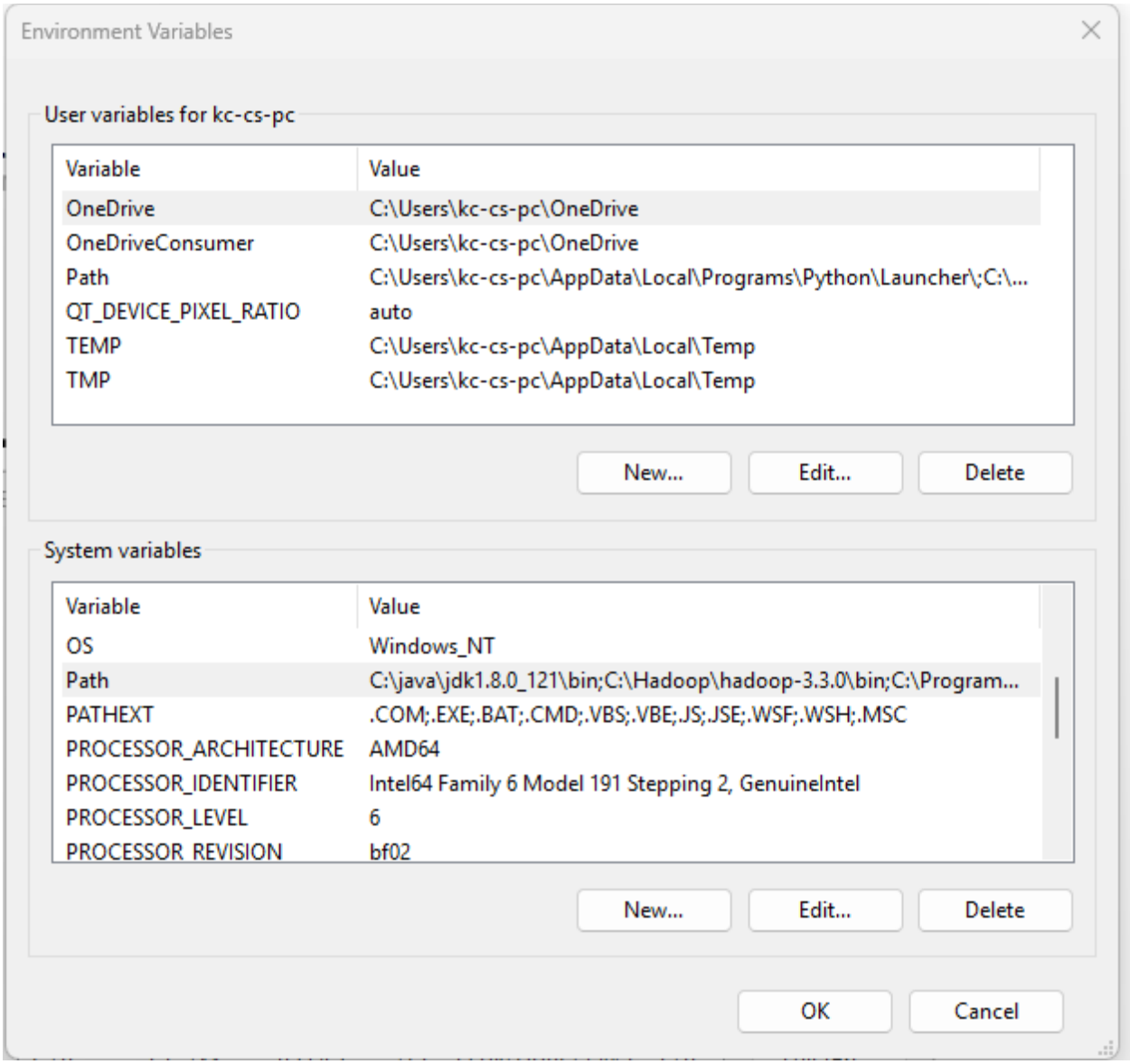
– https://www.apache.org/dyn/closer.cgi/hadoop/common/hadoop 3.3.0/hadoop3.3.0.tar.gz

– extract to C:\Hadoop

Create Hadoop name folder in c n extract in that folder

Set the path HADOOP_HOME Environment variable and set the path till bin





This PC > Windows (C:) >

Sort View ...

Name	Date modified	Type	Size
app	06-10-2025 23:42	File folder	
Hadoop	04-03-2026 09:30	File folder	
inetpub	02-09-2025 09:05	File folder	
java	04-03-2026 09:24	File folder	
ProcLogs	06-10-2025 12:06	File folder	
Program Files	18-02-2026 12:48	File folder	
Program Files (x86)	05-01-2026 14:04	File folder	
shamayla	03-12-2025 13:08	File folder	
TURBOC3	31-10-2025 12:35	File folder	
Users	02-09-2025 09:30	File folder	
Windows	02-03-2026 11:08	File folder	
xampp	27-11-2025 10:28	File folder	
appverifUI.dll	09-07-2025 12:33	Application extens...	110 KB
DumpStack	10-11-2025 11:23	Text Document	12 KB
vfcompat.dll	09-07-2025 12:33	Application extens...	67 KB

Downloads >

Sort View Extract all ...

Name	Date modified	Type	Size
Today			
apache-hadoop-3.1.0-winutils-master	04-03-2026 09:59	Compressed (zipp...	1,062 KB
hadoop-3.3.0.tar.gz	04-03-2026 09:20	PeaZip.GZ	4,89,013 KB
jdk-8u121-windows-x64 (1) (1)	04-03-2026 09:12	Application	2,00,201 KB
BDprac1install (1)	04-03-2026 09:10	Microsoft Word D...	14 KB
BDprac1install	04-03-2026 09:10	Microsoft Word D...	14 KB

3. Do change edit

Configurations

Edit file C:/Hadoop-3.3.0/etc/hadoop/core-site.xml, paste

the xml code in folder and save

```
<configuration>
<property>
<name>fs.defaultFS</name>
<value>hdfs://localhost:9000</value>
</property>
</configuration>
```

==========

Rename “mapred-site.xml.template” to “mapred-site.xml” and edit this file

C:/Hadoop-3.3.0/etc/hadoop/mapred-site.xml, paste xml code and save this file.

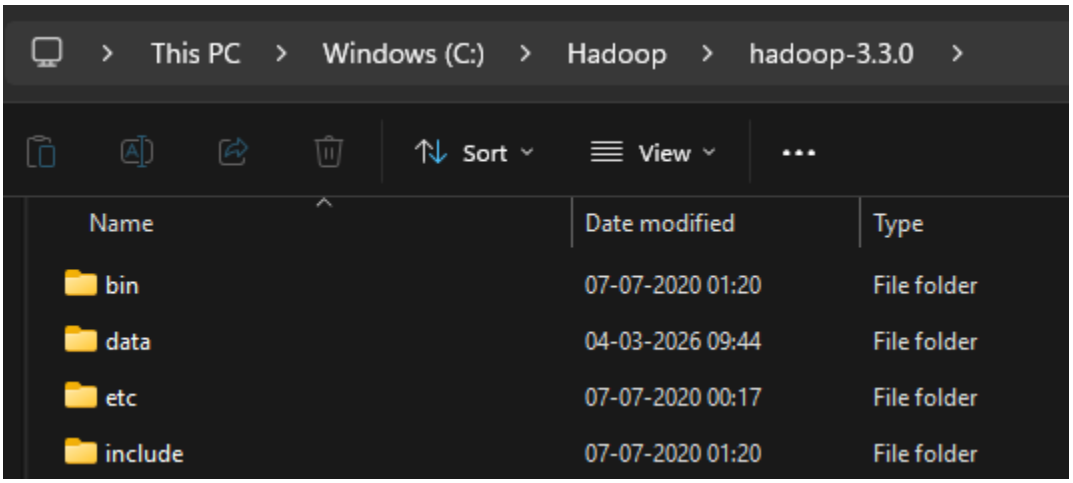
```
<configuration>
<property>
<name>mapreduce.framework.name</name>
<value>yarn</value>
</property>
</configuration>
```

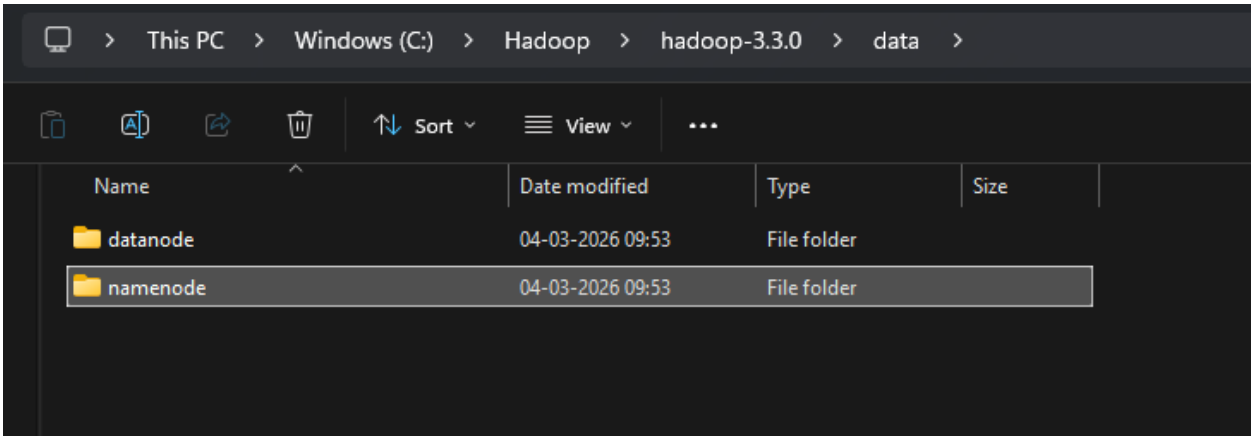
==========

Create folder “data” under “C:\Hadoop-3.3.0”

Create folder “datanode” under “C:\Hadoop-3.3.0\data” Create folder

“namenode” under “C:\Hadoop-3.3.0\data”





==========

Edit file C:\Hadoop-3.3.0/etc/hadoop/hdfs-site.xml, paste xml code

and save this file.

```
<configuration>
<property>
<name>dfs.replication</name>
<value>1</value>
</property>
<property>
<name>dfs.namenode.name.dir</name>
<value>/hadoop-3.3.0/data/namenode</value>
```

```
</property>
<property>
<name>dfs.datanode.data.dir</name>
<value>/hadoop-3.3.0/data/datanode</value>
</property>
</configuration>
```

==========

Edit file C:/Hadoop-3.3.0/etc/hadoop/yarn-site.xml, paste xml code

and save this file.

```
<configuration>
<property>
<name>yarn.nodemanager.aux-services</name>
<value>mapreduce_shuffle</value>
</property>
<property>
<name>yarn.nodemanager.auxservices.mapreduce.shuffle.class</name>
<value>org.apache.hadoop.mapred.ShuffleHandler</value>
</property>
</configuration>
```

==========

Edit file C:/Hadoop-3.3.0/etc/hadoop/hadoop-env.cmd by closing

the command line

“JAVA_HOME=%JAVA_HOME%” instead of set “JAVA_HOME=C:\Java”

Hadoop Configurations Download

https://github.com/brainmentorspvtltd/BigData_RDE/blob/master/Hadoop

%20Configuration.zip or (for hadoop 3)

https://github.com/s911415/apache-hadoop-3.1.0-winutils

– Copy folder bin and replace existing bin folder in

Extract that folder n replace that existing with the new one

Open cmd as admin

C:\Hadoop-3.3.0\bin

– Format the NameNode

– Open cmd and type command “hdfs namenode –format”

Testing

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.26200.7840]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>cd C:\Hadoop\hadoop-3.3.0\bin

C:\Hadoop\hadoop-3.3.0\bin>hdfs namenode -format
2026-03-04 10:34:03,122 INFO namenode.NameNode: STARTUP_MSG:
/************************************************************
STARTUP_MSG: Starting NameNode
STARTUP_MSG:   host = CP-CS-PC38/192.168.56.1
STARTUP_MSG:   args = [-format]
STARTUP_MSG:   version = 3.3.0
STARTUP_MSG:   classpath = C:\Hadoop\hadoop-3.3.0\etc\hadoop;C:\Hadoop\hadoop-3.3.0\share\hadoop\common;C:\Hadoop\hadoop
-3.3.0\share\hadoop\common\lib\accessors-smart-1.2.jar;C:\Hadoop\hadoop-3.3.0\share\hadoop\common\lib\animal-sniffer-ann
otations-1.17.jar;C:\Hadoop\hadoop-3.3.0\share\hadoop\common\lib\asm-5.0.4.jar;C:\Hadoop\hadoop-3.3.0\share\hadoop\commo
n\lib\audience-annotations-0.5.0.jar;C:\Hadoop\hadoop-3.3.0\share\hadoop\common\lib\avro-1.7.7.jar;C:\Hadoop\hadoop-3.3.
0\share\hadoop\common\lib\checker-qual-2.5.2.jar;C:\Hadoop\hadoop-3.3.0\share\hadoop\common\lib\commons-beanutils-1.9.4.
```

– Open cmd and change directory to C:\Hadoop-3.3.0\sbin

– type start-all.cmd

(Or you can start like this)

– Start namenode and datanode with this command

– type start-dfs.cmd

– Start yarn through this command

– type start-yarn.cmd

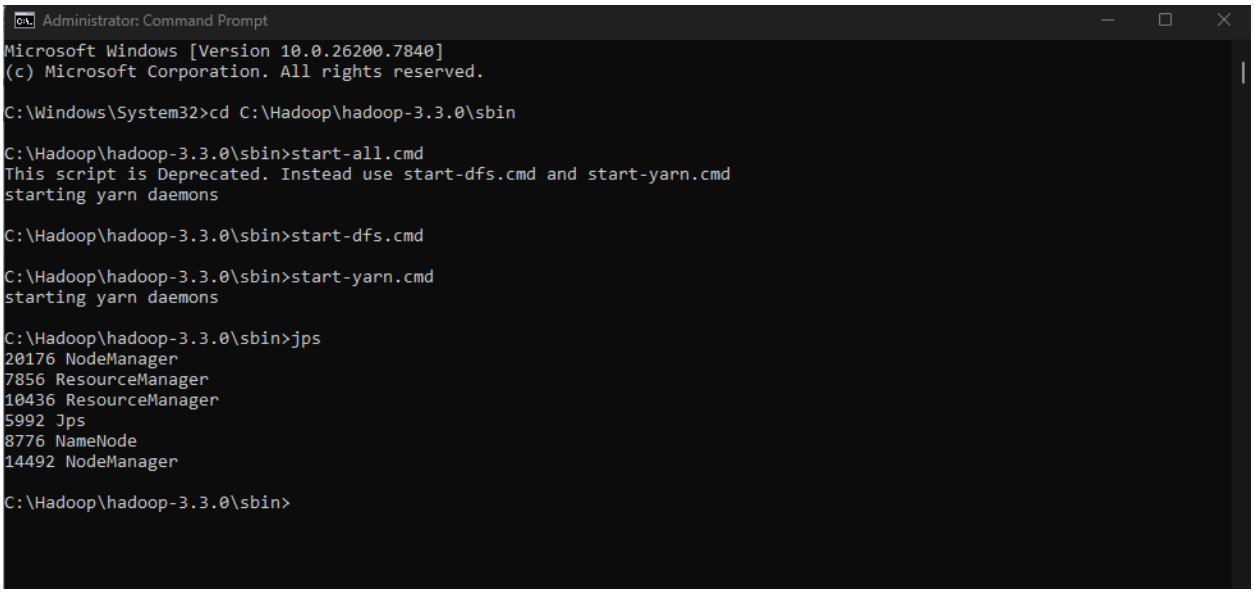
Make sure these apps are running

– Hadoop Namenode

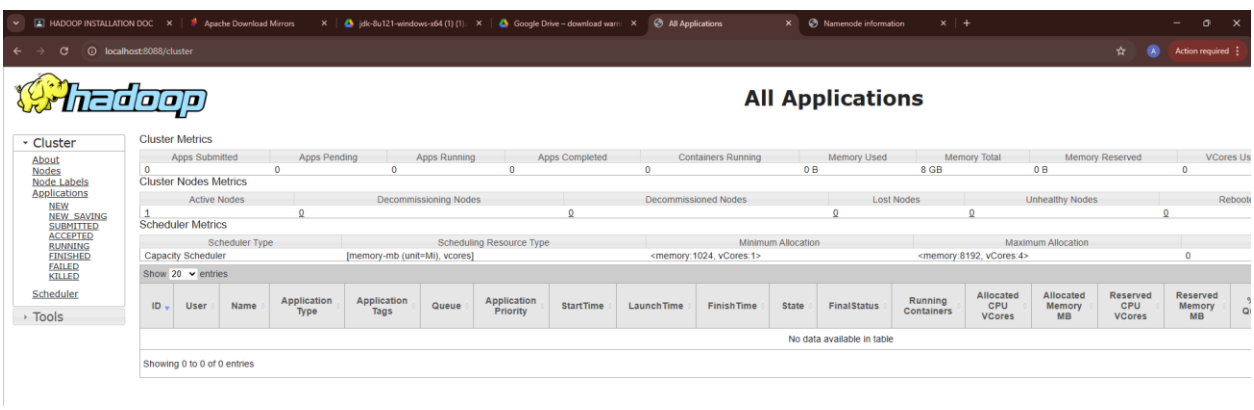
– Hadoop datanode

– YARN Resource Manager

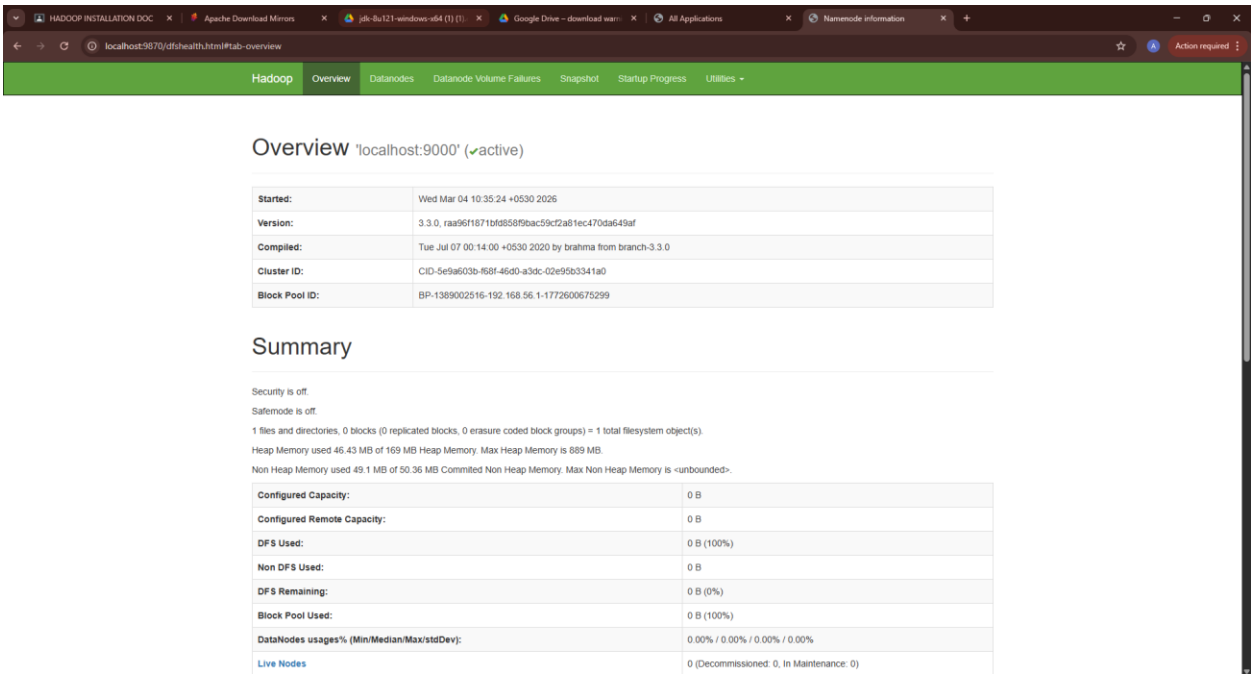
– YARN Node Manager



Open: http://localhost:8088



Open: http://localhost:9870


HADOOP INSTALLATION DOC
Apache Download Mirrors
jdk-8u121-windows-x64 (1) (1)
Google Drive – download warn
All Applications
Namenode information
localhost:9870/dfshealth.html#tab-overview
Action required
Hadoop
Overview
Datanodes
Datanode Volume Failures
Snapshot
Startup Progress
Utilities
Overview 'localhost:9000' (active)
Started: Wed Mar 04 10:35:24 +0530 2026
Version: 3.3.0, raa96f1871bfd858f9bac59cf2a81ec470da649af
Compiled: Tue Jul 07 00:14:00 +0530 2020 by brahma from branch-3.3.0
Cluster ID: CID-5e9a603b-f68f-46d0-a3dc-02e95b3341a0
Block Pool ID: BP-1389002516-192.168.56.1-1772600675299
Summary
Security is off.
Safemode is off.
1 files and directories, 0 blocks (0 replicated blocks, 0 erasure coded block groups) = 1 total filesystem object(s).
Heap Memory used 46.43 MB of 169 MB Heap Memory. Max Heap Memory is 889 MB.
Non Heap Memory used 49.1 MB of 50.36 MB Commited Non Heap Memory. Max Non Heap Memory is <unbounded>.
Configured Capacity: 0 B
Configured Remote Capacity: 0 B
DFS Used: 0 B (100%)
Non DFS Used: 0 B
DFS Remaining: 0 B (0%)
Block Pool Used: 0 B (100%)
DataNodes usages% (Min/Median/Max/stdDev): 0.00% / 0.00% / 0.00% / 0.00%
Live Nodes 0 (Decommissioned: 0, In Maintenance: 0)

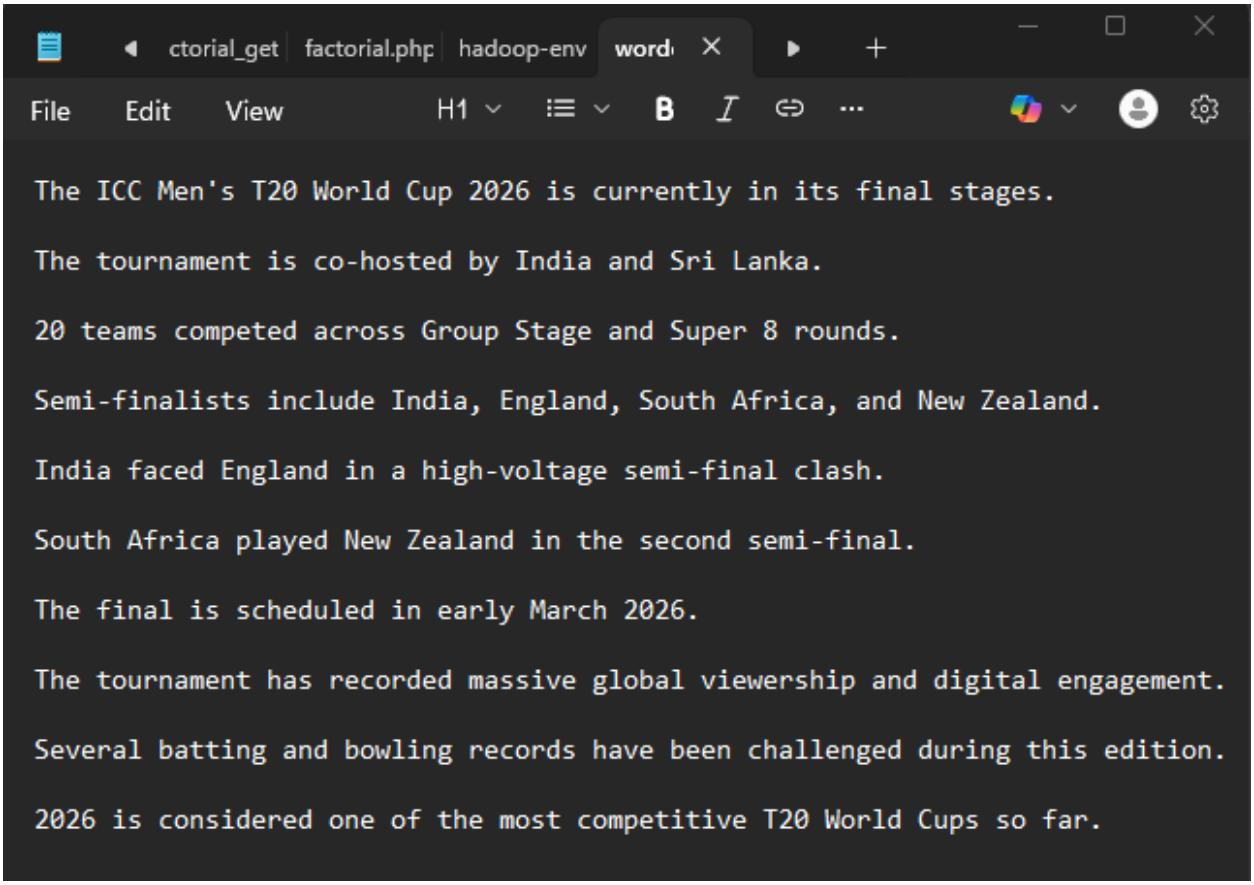
PRACTICAL-2 : MAPREDUCE

STEPS :

First open cmd check check on which path cmd is opening

Go to that path create a text doc there

Write any content & info whatever u want save it



1. **Open idle or any other platform**

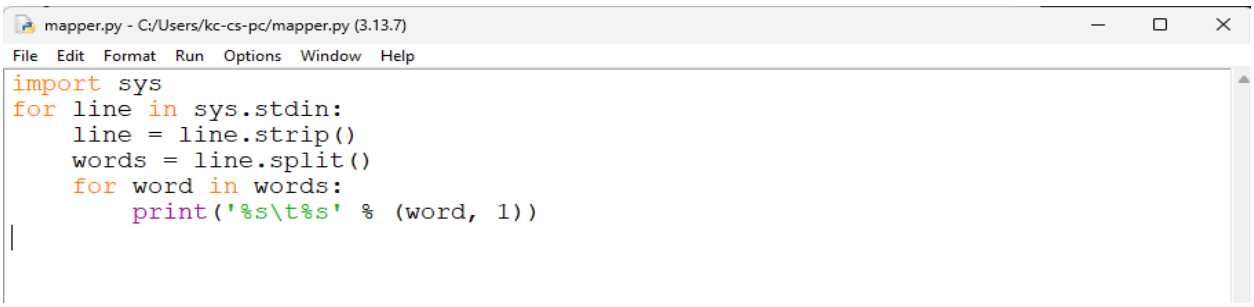
Wrtie this mapper code :

Save this mapper.py where u had saved ur text doc

CODE :

```
import sys
```

```
for line in sys.stdin:
    line = line.strip()
    words = line.split()
    for word in words:
        print('%s\t%s' % (word, 1))
```



Write this recuder code :

Save this reducer.py where u had saved ur text doc

CODE :

```
from operator import itemgetter
import sys
current_word = None
current_count = 0
word = None

for line in sys.stdin:
    line = line.strip()
    word, count = line.split('\t', 1)
    try:
        count = int(count)
    except ValueError:
```

```
        continue
    if current_word == word:
        current_count += count
    else:
        if current_word:
            print('%s\t%s' % (current_word, current_count))
        current_count = count
        current_word = word
if current_word == word:
    print('%s\t%s' % (current_word, current_count))
```

reducer.py - C:/Users/kc-cs-pc/reducer.py (3.13.7)

File Edit Format Run Options Window Help

```
from operator import itemgetter
import sys
current_word = None
current_count = 0
word = None

for line in sys.stdin:
    line = line.strip()
    word, count = line.split('\t', 1)
    try:
        count = int(count)
    except ValueError:
        continue
    if current_word == word:
        current_count += count
    else:
        if current_word:
            print('%s\t%s' % (current_word, current_count))
        current_count = count
        current_word = word
if current_word == word:
    print('%s\t%s' % (current_word, current_count))
```

Open cmd n type this commands :

```
Command Prompt
Microsoft Windows [Version 10.0.26200.7840]
(c) Microsoft Corporation. All rights reserved.

C:\Users\kc-cs-pc>type wordcount.txt
The ICC Men's T20 World Cup 2026 is currently in its final stages.

The tournament is co-hosted by India and Sri Lanka.

20 teams competed across Group Stage and Super 8 rounds.

Semi-finalists include India, England, South Africa, and New Zealand.
```

```
Command Prompt
The tournament has recorded massive global viewership and digital engagement.

Several batting and bowling records have been challenged during this edition.

2026 is considered one of the most competitive T20 World Cups so far.
C:\Users\kc-cs-pc>type mapper.py
import sys
for line in sys.stdin:
    line = line.strip()
    words = line.split()
    for word in words:
        print('%s\t%s' % (word, 1))
```

```
C:\Users\kc-cs-pc>type reducer.py
from operator import itemgetter
import sys
current_word = None
current_count = 0
word = None

for line in sys.stdin:
    line = line.strip()
    word, count = line.split('\t', 1)
    try:
        count = int(count)
    except ValueError:
        continue
    if current_word == word:
        current_count += count
    else:
```

Command Prompt

```
C:\Users\kc-cs-pc>type wordcount.txt|python mapper.py
The     1
ICC     1
Men's   1
T20     1
World   1
Cup     1
2026    1
is      1
currently       1
in      1
its     1
final   1
stages. 1
The     1
```

Command Prompt

```
C:\Users\kc-cs-pc>type wordcount.txt|python mapper.py|sort|python reducer.py
20      1
2026    2
2026.   1
8       1
a       1
across  1
Africa  1
Africa, 1
and     5
batting 1
been    1
bowling 1
by      1
```

PRACTICAL-3 : HBASE

Pre-requisites: We are going to make a standalone setup of HBase in our machine which requires us to

• Install Java JDK 1.8 - We can download and install it from

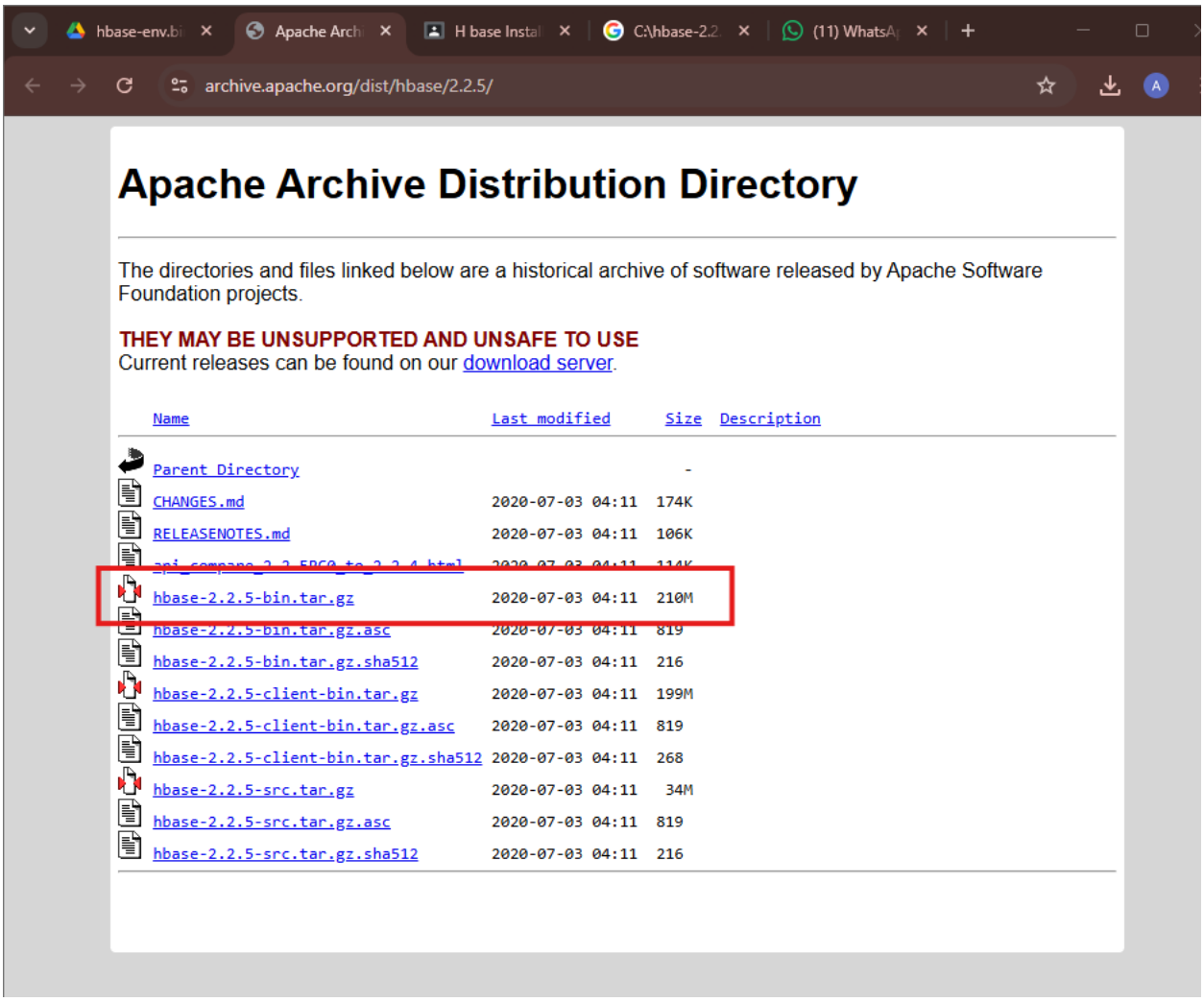
https://www.oracle.com/java/technologies/downloads/

and set JAVA_HOME in

environment variable.

• Download Hbase-bin - Download Apache Hbase-2.2.5 from

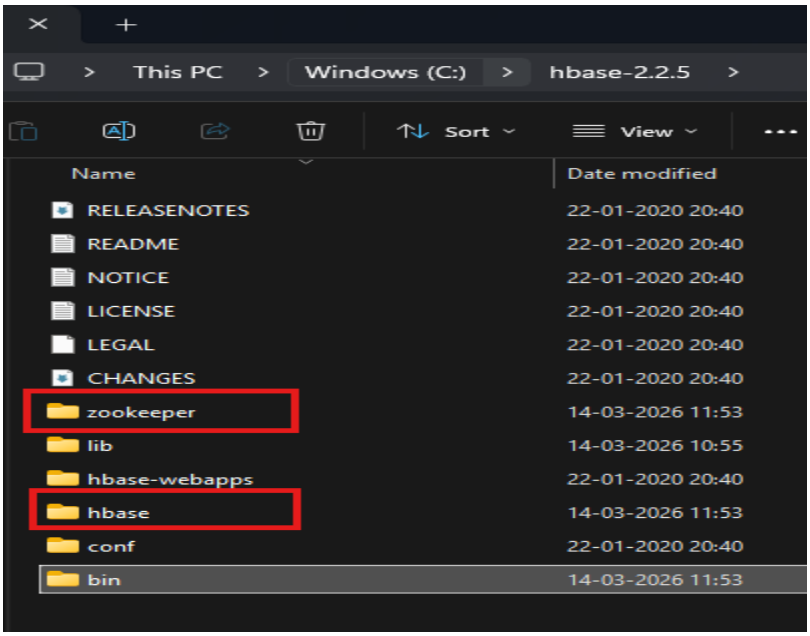
https://archive.apache.org/dist/hbase/2.2.5/



Steps:

• Step 1 – Extract all files from the archive and browse in c drive

• Step 2 – Create folders in cdrive hbase 2.2.5 named "hbase" and "zookeeper"



• Step 3 – Deleting line in HBase.cmd

Open C:\hbase-2.2.5\bin\hbase.cmd in any text editor. Search for line %HEAP_SETTINGS% and remove the highlighted part.

set java_arguments=%HEAP_SETTINGS% %HBASE_OPTS% -classpath "%CLASSPATH%" %CLASS% %hbase-command-arguments%

• Step 4 – Adding lines in hbase-env.cmd

Open C:\hbase-2.2.5\conf\hbase-env.cmd in any text editor. Add the below lines in the file after the comment session.

Hbase.env.bin file download karna hai fir hbase ke conf me jake above path ke file me

set JAVA_HOME=%JAVA_HOME%

set HBASE_CLASSPATH=%HBASE_HOME%\lib\client-facing-thirdparty*

set HBASE_HEAPSIZE=1000

set HBASE_OPTS=-XX:+UseConcMarkSweepGC -Djava.net.preferIPv4Stack=true

set SERVER_GC_OPTS=-verbose:gc -XX:+PrintGCDetails -XX:+PrintGCDateStamps

set HBASE_USE_GC_LOGFILE=true

set HBASE_JMX_BASE=-Dcom.sun.management.jmxremote.ssl=false -Dcom.sun.management.jmxremote.authenticate=false

set HBASE_MASTER_OPTS=%HBASE_JMX_BASE% "-Dcom.sun.management.jmxremote.port=10101"

set HBASE_REGIONSERVER_OPTS=%HBASE_JMX_BASE% "-Dcom.sun.management.jmxremote.port=10102"

set HBASE_THRIFT_OPTS=%HBASE_JMX_BASE% "-Dcom.sun.management.jmxremote.port=10103"

set HBASE_ZOOKEEPER_OPTS=%HBASE_JMX_BASE%-Dcom.sun.management.jmxremote.port=10104"

set HBASE_REGIONSERVERS=%HBASE_HOME%\conf\regionservers

set HBASE_LOG_DIR=%HBASE_HOME%\logs

set HBASE_IDENT_STRING=%USERNAME%

set HBASE_MANAGES_ZK=true

copy kar ke ye paste karna hai

new updated copy-paste

set JAVA_HOME=C:\java\jdk1.8.0_121

```
set HBASE_HOME=C:\hbase-2.2.5

set "HBASE_CLASSPATH=%HBASE_HOME%\lib\client-facing-thirdparty\*"

set HBASE_HEAPSIZE=1000

set HBASE_OPTS=-XX:+UseConcMarkSweepGC -Djava.net.preferIPv4Stack=true

set SERVER_GC_OPTS=-verbose:gc -XX:+PrintGCDetails -XX:+PrintGCDateStamps

set HBASE_USE_GC_LOGFILE=true


set HBASE_JMX_BASE=-Dcom.sun.management.jmxremote.ssl=false -
Dcom.sun.management.jmxremote.authenticate=false


set HBASE_MASTER_OPTS=%HBASE_JMX_BASE% "-
Dcom.sun.management.jmxremote.port=10101"

set HBASE_REGIONSERVER_OPTS=%HBASE_JMX_BASE% "-
Dcom.sun.management.jmxremote.port=10102"

set HBASE_THRIFT_OPTS=%HBASE_JMX_BASE% "-
Dcom.sun.management.jmxremote.port=10103"

set HBASE_ZOOKEEPER_OPTS=%HBASE_JMX_BASE% "-
Dcom.sun.management.jmxremote.port=10104"


set HBASE_REGIONSERVERS=%HBASE_HOME%\conf\regionservers

set HBASE_LOG_DIR=%HBASE_HOME%\logs

set HBASE_IDENT_STRING=%USERNAME%

set HBASE_MANAGES_ZK=true
```

• Step 5 – Adding lines in hbase-site.xml

Open C:\hbase-2.2.5\conf\hbase-site.xml in any text

editor. Add the below lines inside the <configuration>

tag.

```
<property>
```

<name>hbase.rootdir</name>

<value>file:///C:/Documents/hbase-2.2.5/hbase</value>

</property>

<property>

<name>hbase.zookeeper.property.dataDir</name>

<value>/C:/Documents/hbase-2.2.5/zookeeper</value>

</property>

<property>

<name> hbase.zookeeper.quorum</name>

<value>localhost</value>

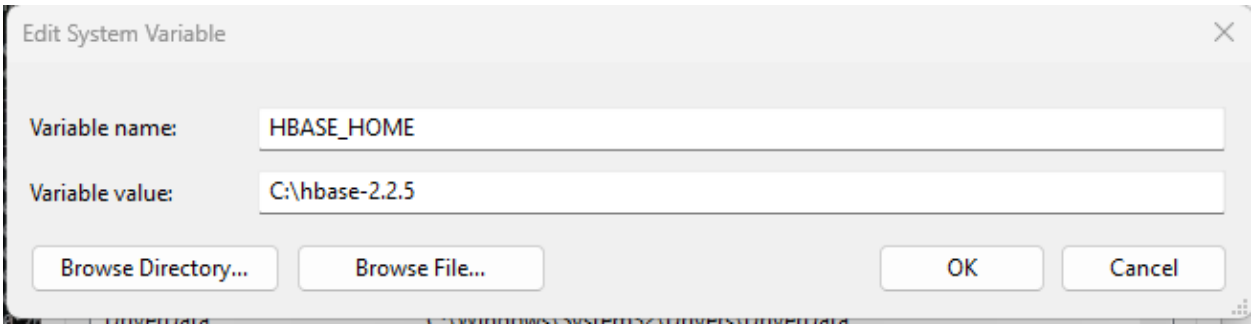
</property>

- **Step 6 – Setting Environment Variables: Input the following:**
- **Variable name: HBASE_HOME**
- **Variable Value: Put the path of the Hbase folder.**

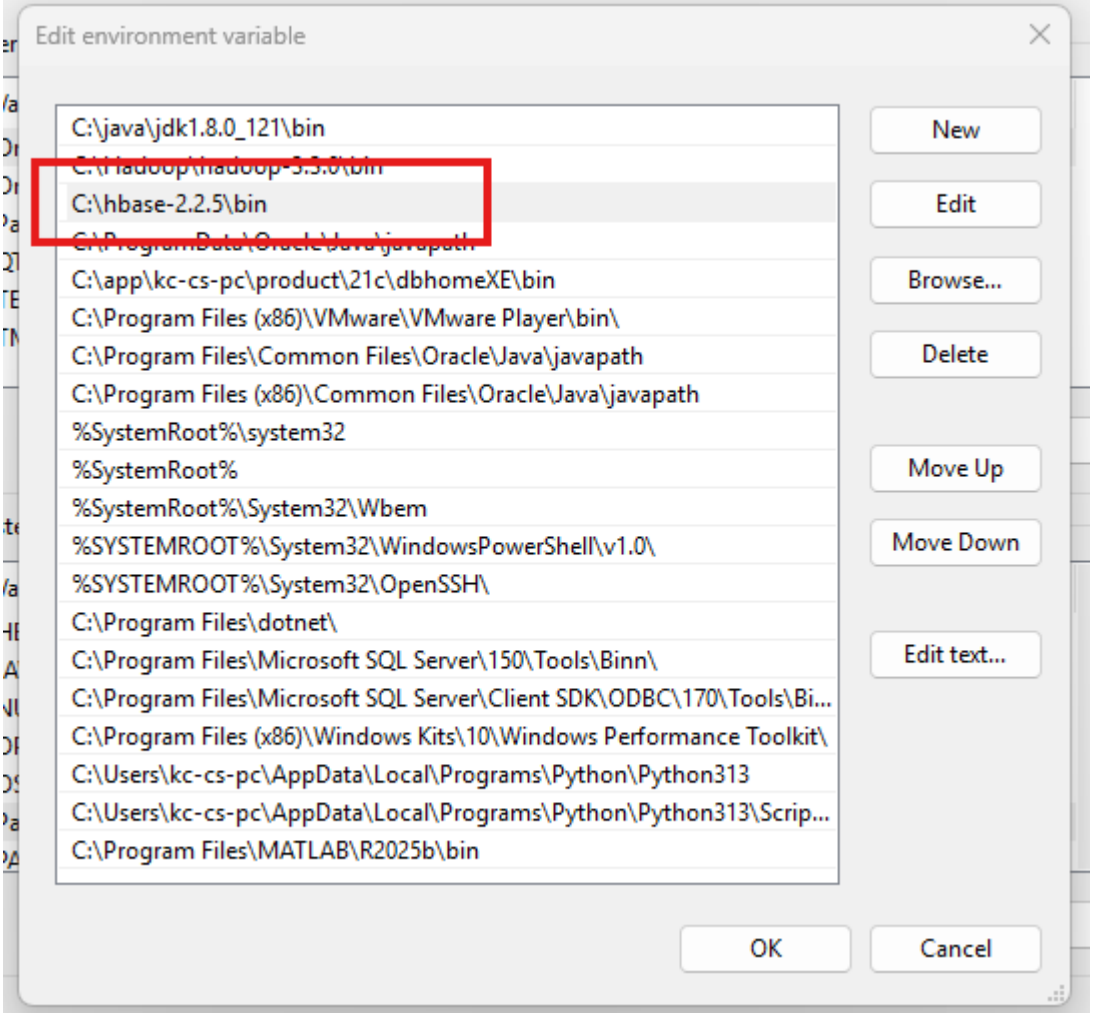
Set the path

HBASE_HOME

C:\hbase-2.2.5



And in path set C:\hbase-2.2.5\bin



Starting the HBASE shell:

• Step 1 –

Use start-hbase.cmd:

Go to the bin folder of hbase and start a cmd prompt

there Then type: start-hbase.cmd

```
Administrator: Command Prompt - hbase  shell
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>cd C:\hbase-2.2.5\bin

C:\hbase-2.2.5\bin>start-hbase.cmd
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/C:/hbase-2.2.5/lib/cl
StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/C:/Hadoop/hadoop-3.3
pl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_binc
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLo
```

- **Step 2 –**

Start hbase shell:

```
C:\hbase-2.2.5\bin>hbase shell
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/C:/hbase-2.2.5/lib/client-fa
StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/C:/Hadoop/hadoop-3.3.0/share
pl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings fo
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFac
HBase Shell
Use "help" to get list of supported commands.
Use "exit" to quit this interactive shell.
```

Status

Syntax: status

This command will give details about the system status like a number of servers present in the cluster, active server count, and average load value. You can also pass any particular parameters depending on how detailed status you want to know about the system. The parameters can be **'summary', 'simple', or 'detailed'**, the default parameter provided is "summary". **Example:**

```
Administrator: Command Prompt - hbase shell
hbase(main):001:0> status
1 active master, 0 backup masters, 1 servers, 0 dead, 2.0000 average load
Took 0.4870 seconds
```

whoami

Syntax:

Syntax: Whoami

This command “whoami” is used to return the current HBase user information from the HBase cluster.

Example:

```
Administrator: Command Prompt - hbase shell
hbase(main):002:0> whoami
kc-cs-pc (auth:SIMPLE)
    groups: ORA_ASMDBA, ORA_DBA, ORA_OraDB21Home1_SYSBACKUP,
ors, Performance Log Users, Users
Took 0.0160 seconds
```

Tables Managements commands

These commands will allow programmers to create tables and table schemas with rows and column families.

The following are Table Management commands

- Create
- List
- Describe
- Disable
- Disable_all
- Enable
- Enable_all
- Drop
- Drop_all
- Show_filters

• Alter

• Alter_status

Let us look into various command usage in HBase with an example.

- **Create**

Syntax: create <tablename>, <columnfamilyname>

Example:

Creating a table named ‘kccs’ with column family ‘students’

```
Administrator: Command Prompt - hbase shell
ors, Performance Log Users, Users
Took 0.0160 seconds
hbase(main):003:0> create 'kccs','students'
Created table kccs
Took 0.8660 seconds
=> Hbase::Table - kccs
```

The above example explains how to create a table in HBase with the specified name given according to the dictionary or specifications as per column family. In addition to this we can also pass some table-scope attributes as well into it. In order to check whether the table ‘kccs’ is created or not, we have to use the “list” command as mentioned below.

- **List**

Syntax:list

```
hbase(main):004:0> list
TABLE
kccs
1 row(s)
Took 0.0320 seconds
=> ["kccs"]
```

• “List” command will display all the tables that are present or created in HBase

• The output showing in above screen shot is currently showing the existing tables in HBase

• We can filter output values from tables by passing optional regular expression parameters

- **Describe**

Syntax:describe <table name>

Example:

```
Administrator: Command Prompt - hbase  shell
=> ["kccs"]
hbase(main):005:0> describe 'kccs'
Table kccs is ENABLED
kccs
COLUMN FAMILIES DESCRIPTION
{NAME => 'students', VERSIONS => '1', EVICT_BLOCKS_ON_CLOSE => 'false', NEW_VERS
ION_BEHAVIOR => 'false', KEEP_DELETED_CELLS => 'FALSE', CACHE_DATA_ON_WRITE => '
false', DATA_BLOCK_ENCODING => 'NONE', TTL => 'FOREVER', MIN_VERSIONS => '0', RE
PLICATION_SCOPE => '0', BLOOMFILTER => 'ROW', CACHE_INDEX_ON_WRITE => 'false', I
N_MEMORY => 'false', CACHE_BLOOMS_ON_WRITE => 'false', PREFETCH_BLOCKS_ON_OPEN =
> 'false', COMPRESSION => 'NONE', BLOCKCACHE => 'true', BLOCKSIZE => '65536'}

1 row(s)

QUOTAS
0 row(s)
Took 0.1590 seconds
```

This command describes the named table.

• It will give more information about column families present in the mentioned table

• In our case, it gives the description about table “kccs”

• It will give information about table name with column families, associated filters, versions and some more details.

- **disable**

Syntax: disable <tablename>

Example: disable 'kccs'

```
Administrator: Command Prompt - hbase  shell
0 row(s)
Took 0.1590 seconds
hbase(main):006:0> disable 'kccs'
Took 0.8110 seconds
hbase(main):007:0> enable 'kccs'
Took 1.3020 seconds
```

- **Enable**

Syntax: enable <tablename>

Example: enable 'kccs’

• This command will start enabling the named table

• Whichever table is disabled, to retrieve back to its previous state we use this command

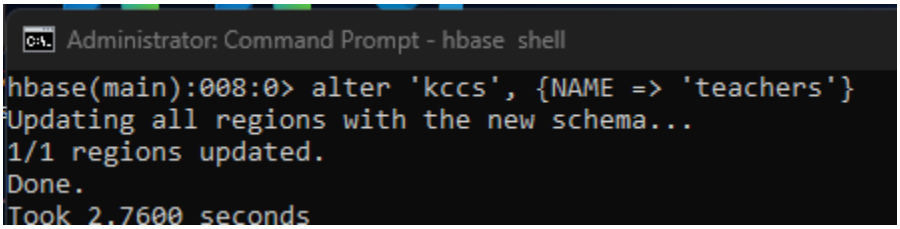
• If a table is disabled in the first instance and not deleted or dropped, and if we want to re-use the disabled table then we have to enable it by using this command.

- **alter**

Syntax: alter <tablename>, NAME=><column familyname>, VERSIONS=>5

This command alters the column family schema. To understand what exactly it does, we have explained it here with an example.

Example: Altering table kccs, by adding a new column family named 'teachers'



Data manipulation commands

These commands will work on the table related to data manipulations such as putting data into a

table, retrieving data from a table and deleting schema, etc.

The commands come under these are

• Count

• Put

• Get

• Delete

• Delete all

• Truncate

• Scan

Let look into these commands usage with an example.

- **Count**

Syntax: count <'tablename'>, CACHE =>1000

• The command will retrieve the count of a number of rows in a table. The value returned by this one is the number of rows.

• Current count is shown per every 1000 rows by default.

• Count interval may be optionally specified.

• Default cache size is 10 rows.

• Count command will work fast when it is configured with right Cache. Example:

```
Administrator: Command Prompt - hbase  shell
Done.
Took 2.7600 seconds
hbase(main):009:0> count 'kccs'
0 row(s)
Took 0.0480 seconds
=> 0
```

- **Put**

Syntax: put <'tablename'>,<'rowname'>,<'columnvalue'>,<'value'>

This command is used for following things

• It will put a cell 'value' at defined or specified table or row or column.

• It will optionally coordinate time stamp. Example:

```
Administrator: Command Prompt - hbase  shell
hbase(main):010:0> put 'kccs',1,'students:name','Amin'
Took 0.0140 seconds
hbase(main):011:0> put 'kccs',1,'students:roll','13'
Took 0.0000 seconds
```

```
Administrator: Command Prompt - hbase  shell
Took 0.0000 seconds
hbase(main):012:0> scan 'kccs'
ROW                    COLUMN+CELL
 1                     column=students:name, timestamp=1773469755742, value=Amin
 1                     column=students:roll, timestamp=1773469770618, value=13
1 row(s)
Took 0.0270 seconds
```

• hbase> put 'kccs',1,'students:name','Dhruv'

Here we are placing values into table “kccs” under row 1, column family ‘students’ and column ‘name’ hbase> put 'kccs',1,'students:roll','59'

Here we are placing values into table “kccs” under row 1, column family ‘students’ and column ‘roll’

• hbase> put 'kccs',1,'teachers:name','Mr. Naveen Pahuja'

Here we are placing values into table “kccs” under row 1, column family ‘teachers’ and column ‘name’

- **Get**

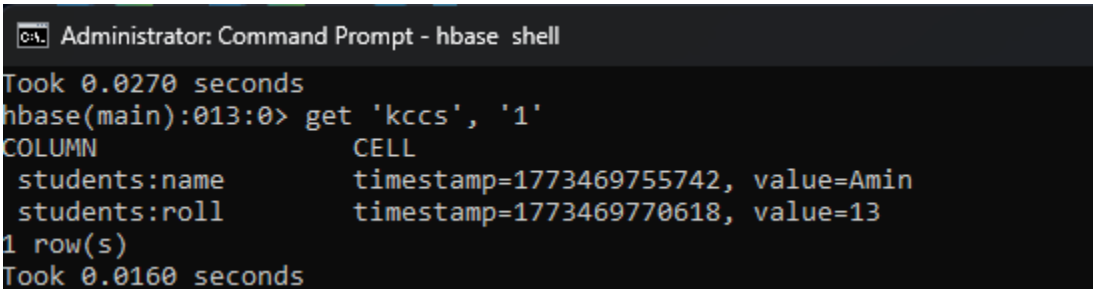
Syntax: get <'tablename'>, <'rowname'>, {< Additional parameters>}

Here <Additional Parameters> include TIMERANGE, TIMESTAMP, VERSIONS and FILTERS.

By using this command, you will get a row or cell contents present in the table. In addition to that

you can also add additional parameters to it like TIMESTAMP, TIMERANGE,VERSIONS,

FILTERS, etc. to get a particular row or cell content.



```
Administrator: Command Prompt - hbase shell
Took 0.0160 seconds
hbase(main):014:0> put 'kccs',1,'teachers:name','Ms. Divya menon'
Took 0.0000 seconds
hbase(main):015:0> scan 'kccs'
ROW                    COLUMN+CELL
 1                     column=students:name, timestamp=1773469755742, value=Amin
 1                     column=students:roll, timestamp=1773469770618, value=13
 1                     column=teachers:name, timestamp=1773469904380, value=Ms. D
                       ivya menon
1 row(s)
Took 0.0000 seconds
```

```
Administrator: Command Prompt - hbase shell
1 row(s)
Took 0.0000 seconds
hbase(main):016:0> get 'kccs', '1', 'teachers'
COLUMN                 CELL
 teachers:name         timestamp=1773469904380, value=Ms. Divya menon
1 row(s)
Took 0.0120 seconds
```

- **Delete**

Syntax:delete <'tablename'>,<'row name'>,<'column name'>

• This command will delete cell value at defined table of row or column.

• Delete must and should match the deleted cells coordinates exactly.

• When scanning, delete cell suppresses older versions of values.

Example: hbase> delete 'kccs',1,'students:roll'

```
Administrator: Command Prompt - hbase shell
Took 0.0120 seconds
hbase(main):017:0> get 'kccs', '1', 'students'
COLUMN                 CELL
 students:name         timestamp=1773469755742, value=Amin
 students:roll         timestamp=1773469770618, value=13
1 row(s)
Took 0.0200 seconds
hbase(main):018:0> delete 'kccs',1,'students:roll'
Took 0.0000 seconds
```

```
Administrator: Command Prompt - hbase  shell
hbase(main):019:0> get 'kccs', '1', 'students'
COLUMN                 CELL
 students:name         timestamp=1773469755742, value=Amin
1 row(s)
Took 0.0100 seconds
```

- **Truncate**

Syntax: truncate <tablename>

After truncate of an hbase table, the schema will present but not the records. This command performs 3 functions; those are listed below

• Disables table if it already presents

• Drops table if it already presents

• Recreates the mentioned table Example:

```
Administrator: Command Prompt - hbase  shell
1 row(s)
Took 0.0100 seconds
hbase(main):020:0> truncate 'kccs'
Truncating 'kccs' table (it may take a while):
Disabling table...
Truncating table...
Took 2.0670 seconds
```

Administrator: Command Prompt - hbase shell

```
ROW                                    COLUMN+CELL
0 row(s)
Took 0.0000 seconds
hbase(main):023:0> get 'kccs', '1', 'students'
COLUMN                                 CELL
0 row(s)
Took 0.0060 seconds
hbase(main):024:0> enable 'kccs'
Took 0.0160 seconds
hbase(main):025:0> get 'kccs', '1', 'students'
COLUMN                                 CELL
0 row(s)
Took 0.0160 seconds
hbase(main):026:0> scan 'kccs'
ROW                                    COLUMN+CELL
0 row(s)
Took 0.0000 seconds
hbase(main):027:0> drop 'kccs'

ERROR: Table kccs is enabled. Disable it first.

For usage try 'help "drop"'

Took 0.0290 seconds
hbase(main):028:0> disable 'kccs'
Took 0.7560 seconds
hbase(main):029:0> drop 'kccs'
Took 0.2710 seconds
hbase(main):030:0> list
TABLE
```

PRACTICAL-4 : PIG

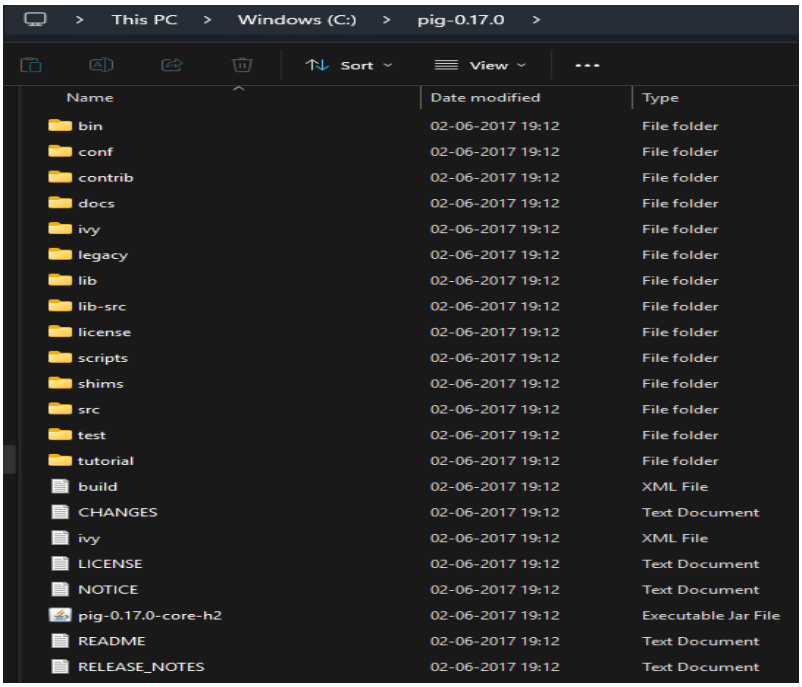
Steps:

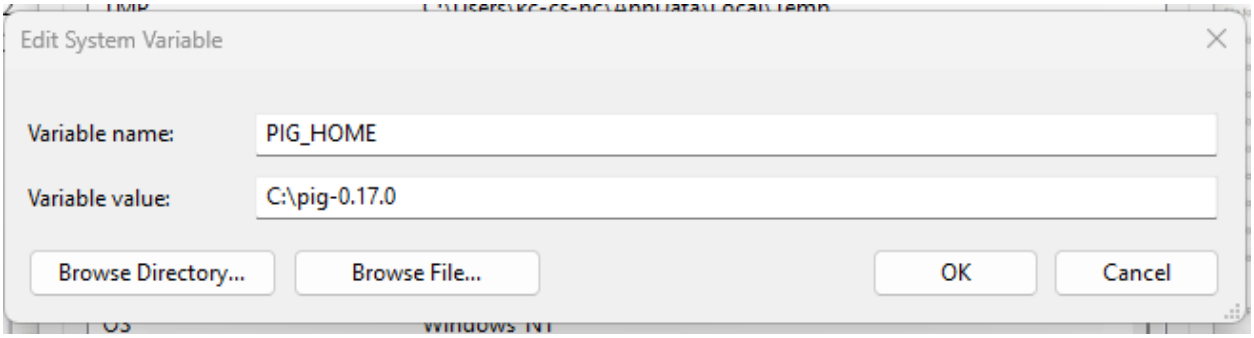
1. Downlaod pig from official page. Download pig-0.17.0.tar.gz file

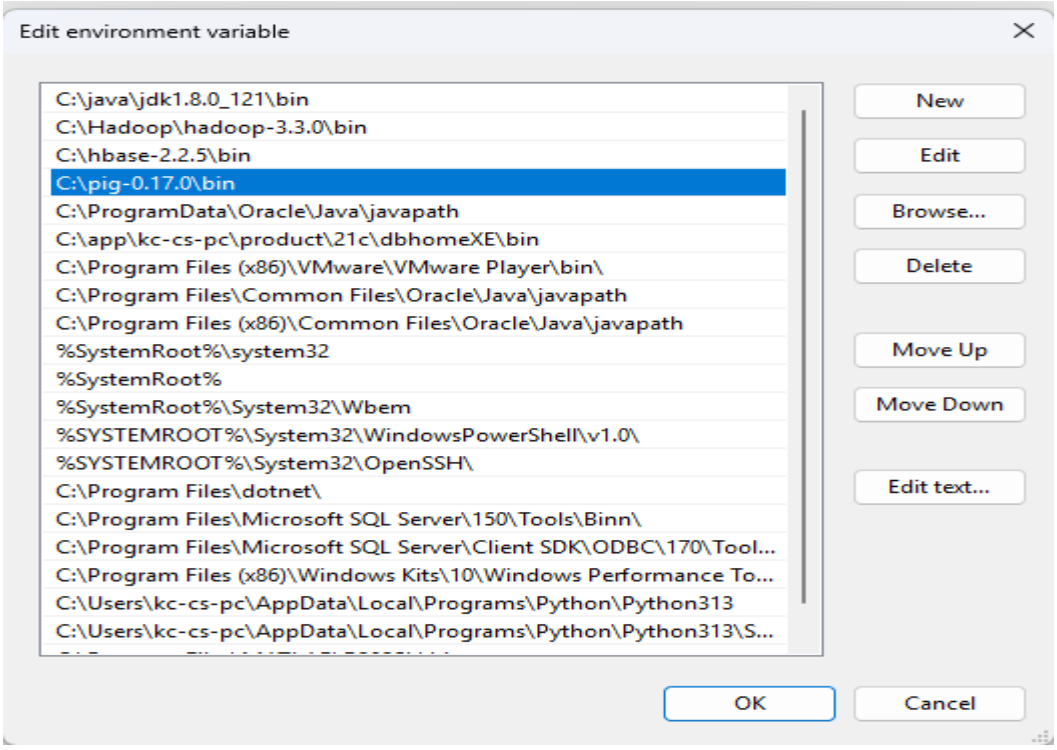
https://downloads.apache.org/pig/pig-0.17.0/

2. Unzip file

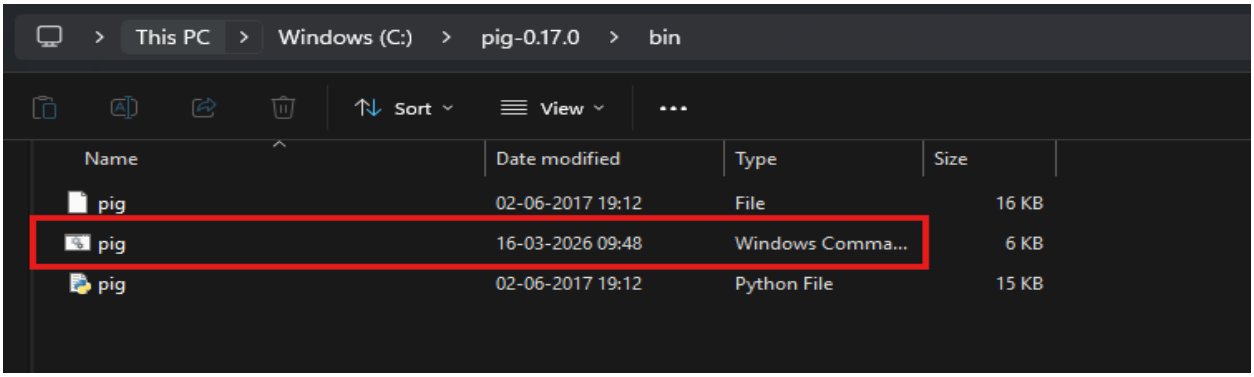
3. Set environment variables

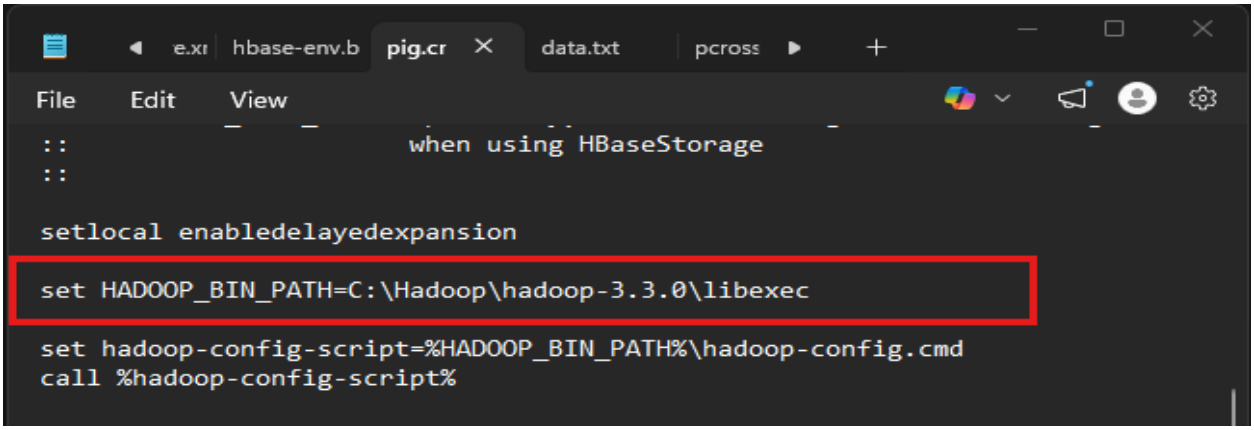






5. If received above error, solve by changing pig.cmd file as below





6. Try pig –version command again.

7. Try running pig in local mode

a. Local mode: pig –x local

Starting Hadoop and pig

Cd C:\hadoop-3.3.1\sbin

Start-all.cmd

Pig –x local

```
Administrator: Command Prompt - pig  -x local;
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>cd C:\Hadoop\hadoop-3.3.0\sbin
```

```
C:\Hadoop\hadoop-3.3.0\sbin>pig -x local;
2026-03-16 09:50:22,364 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-03-16 09:50:22,364 INFO pig.ExecTypeProvider: Picked LOCAL as the ExecType
2026-03-16 09:50:22,519 [main] INFO  org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15
:41:58
2026-03-16 09:50:22,519 [main] INFO  org.apache.pig.Main - Logging error messages to: C:\Hadoop\hadoop-3.3.0\logs\pig_17
73634822512.log
2026-03-16 09:50:22,528 [main] INFO  org.apache.pig.impl.util.Utils - Default bootup file C:\Users\kc-cs-pc/.pigbootup n
ot found
```

- **LOAD AND DUMP:**

A = load ‘data.txt as (lines);

dump A;

```
Administrator: Command Prompt - pig  -x local;
grunt> a = load 'data.txt' as (lines);
2026-03-16 09:50:42,247 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation
ted. Instead, use dfs.bytes-per-checksum
grunt> dump a;
2026-03-16 09:50:46,956 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation
ted. Instead, use dfs.bytes-per-checksum
```

```
Administrator: Command Prompt - pig  -x local;
2026-03-16 09:50:48,056 [main] INFO
o process : 1
(2,1,1,5)
(4,2,9,1)
(5,4,8,3)
(4,7,5,6)
```

DESCRIBE: Describe A;

CROSS PRDOUCT:

C1 = load 'pcross1.txt' using PigStorage(',') as (a1:int,a2:int);

C2 = load 'pcross2.txt' using PigStorage(',') as(a1:int,a2:int,a3:int);

result = cross C1,C2;

dump result;

```
Administrator: Command Prompt - pig  -x local;
grunt> describe a;
a: {lines: bytearray}
grunt> C1 = load 'pcross1.txt' using PigStorage(',') as (a1:int,a2:int);
2026-03-16 10:01:04,127 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation
ted. Instead, use dfs.bytes-per-checksum
grunt> C2 = load 'pcross2.txt' using PigStorage(',') as (a1:int,a2:int,a3:int);
2026-03-16 10:02:48,873 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation
ted. Instead, use dfs.bytes-per-checksum
```

```
Administrator: Command Prompt - pig  -x local;
2026-03-16 10:02:48,873 [main] INFO  org.apache.ha
ted. Instead, use dfs.bytes-per-checksum
grunt> result = cross C1,C2;
grunt> dump result;
```

```
Administrator: Command Prompt - pig  -x local;
s : 1
2026-03-16 10:03:51,768 [main] INFO  org.apache.pig.
o process : 1
(5,9,4,5,3)
(5,9,7,8,9)
(5,9,4,5,6)
(5,9,1,2,3)
(3,4,4,5,3)
(3,4,7,8,9)
(3,4,4,5,6)
(3,4,1,2,3)
(5,2,4,5,3)
(5,2,7,8,9)
(5,2,4,5,6)
(5,2,1,2,3)
(1,2,4,5,3)
(1,2,7,8,9)
(1,2,4,5,6)
(1,2,1,2,3)
```

STORE:

To store the output relation into a folder. Folder will contain another file where the result will be stored

```
Administrator: Command Prompt - pig  -x local;
(1,2,4,5,3)
(1,2,7,8,9)
(1,2,4,5,6)
(1,2,1,2,3)
grunt> store result into 'PigFolder';
2026-03-16 10:06:02,050 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation
ted. Instead, use dfs.bytes-per-checksum
```

```
Administrator: Command Prompt - pig  -x local;

Success!

Job Stats (time in seconds):
JobId   Maps    Reduces MaxMapTime      MinMapTime      AvgMapTime      MedianMapTime
AvgReduceTime   MedianReducetime        Alias   Feature Outputs
job_local758916476_0003 2       1       n/a     n/a     n/a     n/a     n/a     n/a
        file:///C:/Hadoop/hadoop-3.3.0/sbin/PigFolder,

Input(s):
Successfully read 4 records from: "file:///C:/Hadoop/hadoop-3.3.0/sbin/pcross2.txt"
Successfully read 4 records from: "file:///C:/Hadoop/hadoop-3.3.0/sbin/pcross1.txt"

Output(s):
Successfully stored 16 records in: "file:///C:/Hadoop/hadoop-3.3.0/sbin/PigFolder"

Counters:
Total records written : 16
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0
```

FILTER: Will filter input based on a criteria provided.

b = load 'filter.txt' using PigStorage(',') as (a1:int,a2:int);

dump b;

result = filter b by a2==8;

dump result;

```
Administrator: Command Prompt - pig  -x local;
2026-03-16 10:07:12,656 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapRe
ccess!
grunt> b = load 'filter.txt' using PigStorage(',') as (a1:int,a2:int);
2026-03-16 10:12:04,662 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation -
ted. Instead, use dfs.bytes-per-checksum
2026-03-16 10:12:04,678 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation -
ted. Instead, use dfs.bytes-per-checksum
grunt> dump b;
2026-03-16 10:12:13,871 [main] INFO  org.apache.pig.tools.pigstats.ScriptState - Pig fea
N
```

```
Administrator: Command Prompt - pig  -x local;
o process : 1
(1,2)
(4,5)
(7,8)
(3,6)
```

```
Administrator: Command Prompt - pig  -x local;
(7,8)
(3,6)
grunt> result = filter b by a2==8;
grunt> dump result;
2026-03-16 10:12:53,131 [main] INFO  org.apache.pig.tools.pigstats.ScriptState -
```

```
Administrator: Command Prompt - pig  -x local;
2026-03-16 10:12:53,523 [main] INFO
s : 1
2026-03-16 10:12:53,523 [main] INFO
o process : 1
(7,8)
```

FOREACH: Generates data transformations based on columns of data.

a = load 'foreach.txt' using PigStorage(',') as (a1:int,a2:int,a3:int);

dump a;

fe = foreach a generate a1,a2;

dump fe;

```
Administrator: Command Prompt - pig  -x local;
o process : 1
(7,8)
grunt> a = load 'foreach.txt' using PigStorage(',') as (a1:int,a2:int,a3:int);
2026-03-16 10:16:12,063 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation -
ted. Instead, use dfs.bytes-per-checksum
grunt> dump a;
2026-03-16 10:16:16,522 [main] INFO  org.apache.pig.tools.pigstats.ScriptState - Pig fea
N
```

```
Administrator: Command Prompt - pig  -x local;
2026-03-16 10:16:16,905 [main] INFO  org.
o process : 1
(1,2,3)
(4,5,6)
(7,8,9)
```

```
Administrator: Command Prompt - pig  -x local;
(4,5,6)
(7,8,9)
grunt> fe = foreach a generate a1,a2;
grunt> dump fe;
2026-03-16 10:16:50,199 [main] INFO  org.apache.pig.tools.pigstats.ScriptState -
N
```

```
Administrator: Comma
2026-03-16 10:16:56
s : 1
2026-03-16 10:16:56
o process : 1
(1,2)
(4,5)
(7,8)
```

GROUP:

Group operator is used to group the data in one or more relations. It groups the tuples that contain a similar group key. If the group key has more than one field, it treats it as tuple otherwise it will be the same type as that of the group key. In a result, it provides a relation that contains one tuple per group.

A = load 'input.txt' using PigStorage(',') as (fname:chararray,lname,chararray,id:int);

dump A;

groupFname = group A by fname; dump

groupFname;

```
grunt> A = load 'input.txt' using PigStorage(',') as (fname:chararray,lname,chararray,id:int);
2026-03-16 10:27:56,164 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - io.bytes.pe
ted. Instead, use dfs.bytes-per-checksum
grunt> dump A;
2026-03-16 10:28:02,974 [main] INFO  org.apache.pig.tools.pigstats.ScriptState - Pig features used
N
```

```
Administrator: Command Prompt - pig  -x local;
s : 1
2026-03-16 10:28:03,371 [main] INFO  org.apache
o process : 1
((Amin,Khan,1,)
((Rehan,Mujawar,2,)
((Ahmed,Rumaney,3,)
((Shakir,Ansari,4,)
((Juveria,Ansari,5,)
((Alma,Khan,6,)
```

```
Administrator: Command Prompt - pig  -x local;
grunt> groupFname = group A by fname;
grunt> dump groupFname;
2026-03-16 10:29:04,233 [main] INFO  org.apac
BY
```

```
Administrator: Command Prompt - pig  -x local;
s : 1
2026-03-16 10:29:04,771 [main] INFO  org.a
o process : 1
((Alma,{((Alma,Khan,6,)})
((Amin,{((Amin,Khan,1,)})
((Ahmed,{((Ahmed,Rumaney,3,)})
((Rehan,{((Rehan,Mujawar,2,)})
((Shakir,{((Shakir,Ansari,4,)})
((Juveria,{((Juveria,Ansari,5,)})
```

LIMIT:

Limit no of output tuples

l = load 'limit.txt' using PigStorage(',') as (a1:int,a2:int,a3:int);

dump l;

result = limit l 2;

dump result;

```
Administrator: Command Prompt - pig  -x local;
grunt> l = load 'limit.txt' using PigStorage(',') as (a1:int,a2:int,a3:int);
2026-03-16 10:32:11,313 [main] INFO  org.apache.hadoop.conf.Configuration.depre
ted. Instead, use dfs.bytes-per-checksum
grunt> dump l;
2026-03-16 10:32:23,488 [main] INFO  org.apache.pig.tools.pigstats.ScriptState
N
```

```
Administrator: Command Prompt
o process : 1
(,2,)
(,5,)
(,8,)
(,6,)
```

```
grunt> result = limit l 2;
grunt> dump result;
2026-03-16 10:33:35,425 [main] INFO  o
```

```
Administrator: Command Prompt
s : 1
2026-03-16 10:33:35,513 [m
o process : 1
(,2,)
(,5,)
grunt> o = load 'order.txt
```

ORDER BY:

The Apache Pig ORDER BY operator sorts a relation based on one or more fields. It

maintains the order of tuples

o = load 'order.txt' using PigStorage(',') as (a1:int,a2:int,a3:int);

dump o;

result = order o by a1 desc;

Dump result;

```
Administrator: Command Prompt - pig  -x local;
(,2,)
(,5,)
grunt> o = load 'order.txt' using PigStorage(',') as (a1:int,a2:int,a3:int);
2026-03-16 10:37:22,756 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation -
ted. Instead, use dfs.bytes-per-checksum
grunt> dump o;
2026-03-16 10:37:27,641 [main] INFO  org.apache.pig.tools.pigstats.ScriptState - Pig fe
```

```
Administrator: Command Prompt - pig  -x local;
2026-03-16 10:37:28,012 [main] INFO  org
o process : 1
(1,2,3)
(4,5,6)
(7,8,9)
(3,6,9)
```

```
Administrator: Command Prompt - pig  -x local;
Details at logfile: C:\Hadoop\hadoop-3.3.0\logs
grunt> result = order o by a1 desc;
grunt> dump result;
2026-03-16 10:38:18,270 [main] INFO  org.apache
BY
```

```
Administrator: Command Promp
s : 1
2026-03-16 10:38:19,593 [
o process : 1
(7,8,9)
(4,5,6)
(3,6,9)
(1,2,3)
```

SPLIT:

The Apache Pig SPLIT operator breaks the relation into two or more relations according to the provided expression. Here, a tuple may or may not be assigned to one or more than one relation.

s = load 'split.txt' using PigStorage(',') as (a1:int,a2:int);

dump s;

split s into x if a1<=2, y if a1>2;

dump x;

```
Administrator: Command Prompt - pig  -x local;
(1,2,3)
grunt> s = load 'split.txt' using PigStorage(',') as (a1:int,a2:int);
2026-03-16 10:40:20,395 [main] INFO  org.apache.hadoop.conf.Configuration.
ted. Instead, use dfs.bytes-per-checksum
grunt> dump s;
2026-03-16 10:40:32,788 [main] INFO  org.apache.pig.tools.pigstats.ScriptS
```

```
Administrator: Command Promp
2026-03-16 10:40:33,157 [
s : 1
2026-03-16 10:40:33,157 [
o process : 1
(1,2)
(4,5)
(7,8)
(3,6)
(9,8)
```

```
Administrator: Command Prompt - pig  -x local;
grunt> split s into x if a1<=2, y if a1>2;
grunt> dump x;
2026-03-16 10:41:31,235 [main] INFO  org.apache.pig.tools.
N
```

```
Administrator: Command Prompt - pig -
o process : 1
(1,2)
```

DISTINCT:

Used to remove duplicate tuples in a relation. Pig sorts the data and then eliminated

duplicates

d = load 'distinct.txt' using PigStorage(',') as (a1:int,a2:int,a3:int);

dump d;

result = distinct d;

dump result;

```
Administrator: Command Prompt - pig  -x local;
o process : 1
(1,2)
grunt> d = load 'distinct.txt' using PigStorage(',') as (a1:int,a2:int,a3:int);
2026-03-16 10:44:11,945 [main] INFO  org.apache.hadoop.conf.Configuration.deprecat
ted. Instead, use dfs.bytes-per-checksum
grunt> dump d;
2026-03-16 10:44:17,208 [main] INFO  org.apache.pig.tools.pigstats.ScriptState - P
N
```

```
Administrator: Command Prompt - pig  -x local;
2026-03-16 10:44:17,591 [main] INFO  org.apache.pig.backend.
o process : 1
(1,2,3)
(4,5,6)
(7,8,9)
(3,6,9)
(7,5,3)
grunt> result = distinct d;
grunt> dump result;
```

```
Administrator: Command Prompt - pi
2026-03-16 10:44:36,208 [mair
o process : 1
(1,2,3)
(3,6,9)
(4,5,6)
(7,5,3)
(7,8,9)
```

JOIN:

The JOIN operator is used to combine records from two or more relations. While performing a join operation, we declare one (or a group of) tuple(s) from each relation, as keys. When these keys match, the two particular tuples are matched, else the records are dropped. Joins can be of the following types:

a. Self Join : Used to join a table with itself, as if there are 2 relations, temporarily renaming at least 1 relation

a = load 'customers.txt' using PigStorage(',') as

(id:int,name:chararray,age:int,address:chararray,salary:int);

b = load 'customers.txt' using PigStorage(',') as

(id:int,name:chararray,age:int,address:chararray,salary:int);

result = join a by id, b by id;

dump result;

```
grunt> a = load 'customers.txt' using PigStorage(',') as (id:int,name:chararray,age:int,address:chararray,salary:int);
2023-04-08 10:38:10,648 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - io.bytes.per.checksum is depreca
ted. Instead, use dfs.bytes-per-checksum
grunt> b = load 'customers.txt' using PigStorage(',') as (id:int,name:chararray,age:int,address:chararray,salary:int);
2023-04-08 10:39:43,093 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - io.bytes.per.checksum is depreca
ted. Instead, use dfs.bytes-per-checksum
grunt> result = join a by id, b by id;
grunt> dump result;
```

b. Inner Join : Also referred to as Equijoin. Returns rows if there is a match in both tables.

a = load 'customers.txt' using PigStorage(',') as (id:int,name:chararray,age:int,address:chararray,salary:int);

b = load 'orders.txt' using PigStorage(',') as (oid:int,date:chararray,customer_id:int,amount:int);

c = join a by id, b by customer_id;

dump c;

```
grunt> a = load 'customers.txt' using PigStorage(',') as (id:int,name:chararray,age:int,address:chararray,salary:int);
2023-04-08 10:42:32,823 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - io.bytes.per.checksum is deprecated. Instead, use dfs.bytes-per-checksum
grunt> b = load 'orders.txt' using PigStorage(',') as (oid:int,date:chararray,customer_id:int,amount:int);
2023-04-08 10:43:58,474 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - io.bytes.per.checksum is deprecated. Instead, use dfs.bytes-per-checksum
grunt> c = join a by id, b by customer_id;
grunt> dump c;
```

```
o process : 1
(2,Khilan,25,Delhi,1500,101,2009-11-20 00:00:00,2,1500)
(3,kaushik,23,Kota,2000,100,2009-10-08 00:00:00,3,1500)
(3,kaushik,23,Kota,2000,102,2009-10-08 00:00:00,3,3000)
(4,Bharat,25,Mumbai,6500,103,2008-05-20 00:00:00,4,2000)
grunt>
```

c. Outer Join : Returns all rows from at least one of the relations. Can be Left outer join, Right outer join, or full outer join.

1. Left outer join : Returns all rows from the left table, even if there are no matches in the right relation.

2. Right outer join : Returns all rows from the right table, even if there are no matches in the left table

3. Full Outer Join : Returns all rows when there is a match in match in one of the relations

. d = join a by id LEFT OUTER, b by customer_id;

dump d;

```
grunt> d = join a by id LEFT OUTER, b by customer_id;
grunt> dump d;
```

```
(1,Ramesh,32,Ahmedabad,2000,,,,)
(2,Khilan,25,Delhi,1500,101,2009-11-20 00:00:00,2,1500)
(3,kaushik,23,Kota,2000,100,2009-10-08 00:00:00,3,1500)
(3,kaushik,23,Kota,2000,102,2009-10-08 00:00:00,3,3000)
(4,Bharat,25,Mumbai,6500,103,2008-05-20 00:00:00,4,2000)
(5,Sachin.27.Bhopal,8500,,,,,,)
(6,Komal,22,MP,4500,,,,)
(7,Muffy,24,Indore,10000,,,,)
grunt>
```

e = join a by id Right, b by customer_id;

dump e;

```
grunt> e = join a by id Right, b by customer_id;
grunt> dump e;
```

```
(2,Khilan,25,Delhi,1500,101,2009-11-20 00:00:00,2,1500)
(3,kaushik,23,Kota,2000,100,2009-10-08 00:00:00,3,1500)
(3,kaushik,23,Kota,2000,102,2009-10-08 00:00:00,3,3000)
(4,Bharat,25,Mumbai,6500,103,2008-05-20 00:00:00,4,2000)
grunt>
```

f = join a by id full outer, b by customer_id;

```
grunt> f = join a by id full outer, b by customer_id;
grunt> dump f;
```

```
o process : 1
(1,Ramesh,32,Ahmedabad,2000,,,,)
(2,Khilan,25,Delhi,1500,101,2009-11-20 00:00:00,2,1500)
(3,kaushik,23,Kota,2000,100,2009-10-08 00:00:00,3,1500)
(3,kaushik,23,Kota,2000,102,2009-10-08 00:00:00,3,3000)
(4,Bharat,25,Mumbai,6500,103,2008-05-20 00:00:00,4,2000)
(5,Sachin.27.Bhopal,8500,,,,,,)
(6,Komal,22,MP,4500,,,,)
(7,Muffy,24,Indore,10000,,,,)
grunt>
```

Example: Working with a csv file

1. Creating the csv file containing sales records

2. Load the records and group them based on the city

```
sales = load 'sales.csv' using PigStorage(',') as
(tdate:chararray,product:chararray,price:chararray,ptype:chararray,city:chararray);
dump sales;
groupByCity = group sales by city;
dump groupByCity;
```

```
grunt> sales = load 'sales.csv' using PigStorage(',') as (tdate:chararray,product:chararray,price:chararray,ptype:charar
ray,city:chararray);
2023-04-08 10:56:23,980 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - io.bytes.per.checksum is depreca
ted. Instead, use dfs.bytes-per-checksum
grunt> dump sales:
```

```
('12-03-2020', 'shirts','250','Cash', 'Joe')
('11-12-2022', 'trousers', '450','Card', 'John')
('19-67-2021', 'backpack', '1600', 'Cash', 'John')
('125-609-2023', 'suitcase', '2560', 'Card', 'Neo')
('21-10-2019', 'bottle', '100', 'Card', 'Michael')
('31-01-2017', 'soap', '50', 'Cash','Jin')
('129-085-2020', 'gloves', '150','Card', 'Dwight')
('128-062-2022','telephone', '500','Cash', 'Calvin')
('09-11-2018', 'batteries','40', 'Cash', 'Dwayne')
('117-808-2019', 'tie','140','Card', 'Adam')
grunt> groupByCity = group sales by city;
grunt> dump groupByCity;
```

```
('Jin',{('31-01-2017', 'soap', '50', 'Cash','Jin')})
('Joe',{('12-03-2020', 'shirts','250','Cash', 'Joe')})
('Neo',{('125-609-2023', 'suitcase', '2560', 'Card', 'Neo')})
('Adam',{('117-808-2019', 'tie','140','Card', 'Adam')})
('John',{('19-67-2021', 'backpack', '1600', 'Cash', 'John'),('11-12-2022', 'trousers', '450','Card', 'John')})
('Calvin',{('128-062-2022','telephone', '500','Cash', 'Calvin')})
('Dwayne',{('09-11-2018', 'batteries','40', 'Cash', 'Dwayne')})
('Dwight',{('129-085-2020', 'gloves', '150','Card', 'Dwight')})
('Michael',{('21-10-2019', 'bottle', '100', 'Card', 'Michael')})
grunt>
```

3. Count the number of sales in each city.

Note: Concat command will concatenate 2 or more fields of the same type

```
countByCity = foreach groupByCity GENERATE
CONCAT((chararray)$0,CONCAT(':',(chararray)COUNT($1)));
dump f;
```

4. Store the sales records obtained earlier into a folder and verify them

```
store countByCity into 'pig_output_sales' using PigStorage('\t');
```

```
grunt> countByCity = foreach groupByCity GENERATE CONCAT((chararray)$0,CONCAT(':',(chararray)CO
grunt> dump countByCity;
```

```
o process : 1
('Jin':1)
( 'Joe':1)
( 'Neo':1)
( 'Adam':1)
( 'John':2)
( 'Calvin':1)
( 'Dwayne':1)
( 'Dwight':1)
( 'Michael':1)
grunt>
```

4. **Store the sales records obtained earlier into a folder and verify them**

   store countByCity into 'pig_output_sales' using PigStorage('\t');

```
grunt> store countByCity into 'pig_output_sales' using PigStorage('\t');
```

```
job_local217405353_0032 1       1       n/a     n/a     n/a     n/a     n/a     n/
pByCity,sales   GROUP_BY,COMBINER       file:///C:/hadoop-3.3.1/sbin/pig_output_sa

Input(s):
Successfully read 10 records from: "file:///C:/hadoop-3.3.1/sbin/sales.csv"

Output(s):
Successfully stored 9 records in: "file:///C:/hadoop-3.3.1/sbin/pig_output_sales"

Counters:
Total records written : 9
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
```

PC › Local Disk (C:) › hadoop-3.3.1 › sbin › pig_output_sales

Name	Date modified
._SUCCESS.crc	08-04-2023 11:13 AM
.part-r-00000.crc	08-04-2023 11:13 AM
_SUCCESS	08-04-2023 11:13 AM
part-r-00000	08-04-2023 11:13 AM